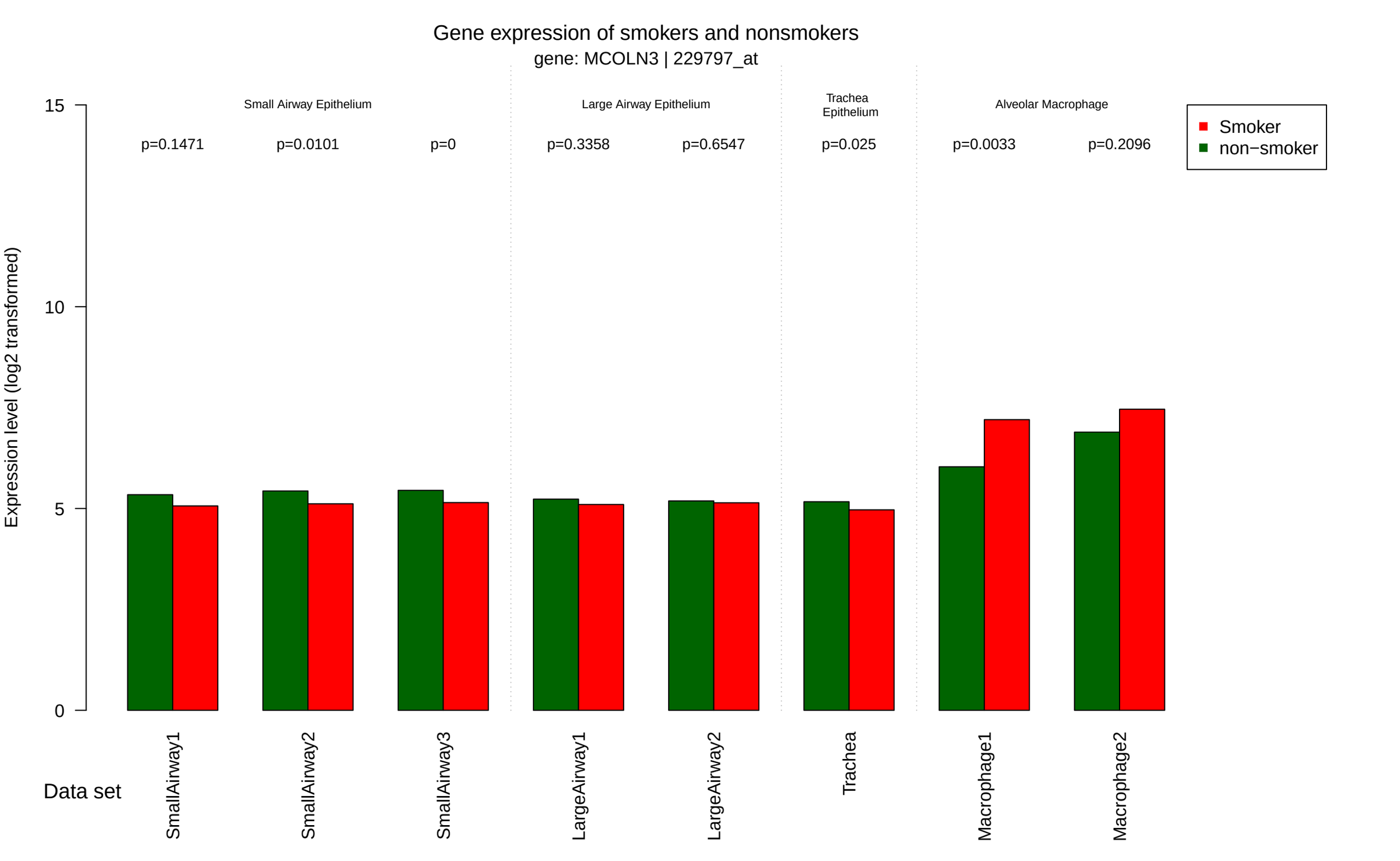
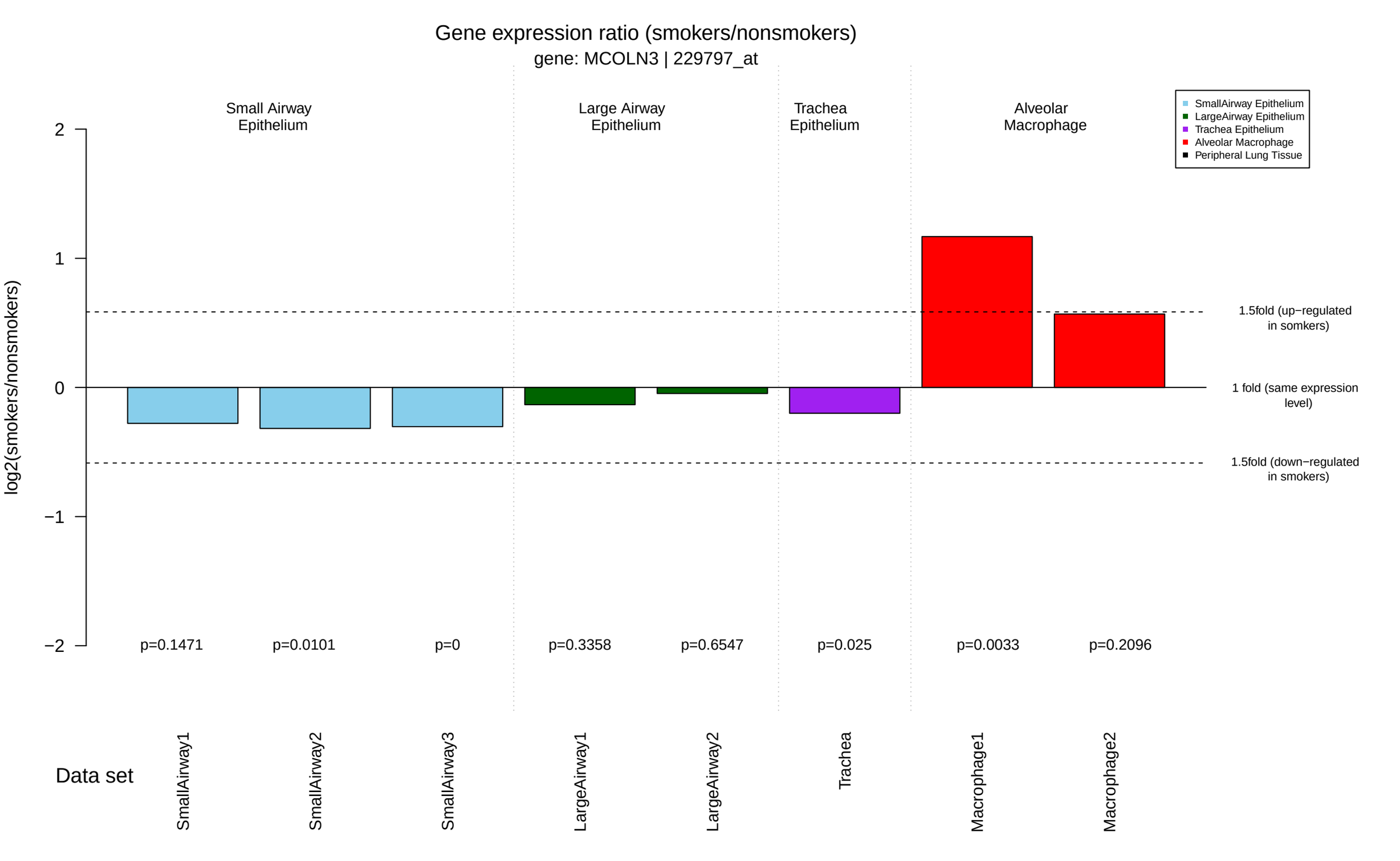
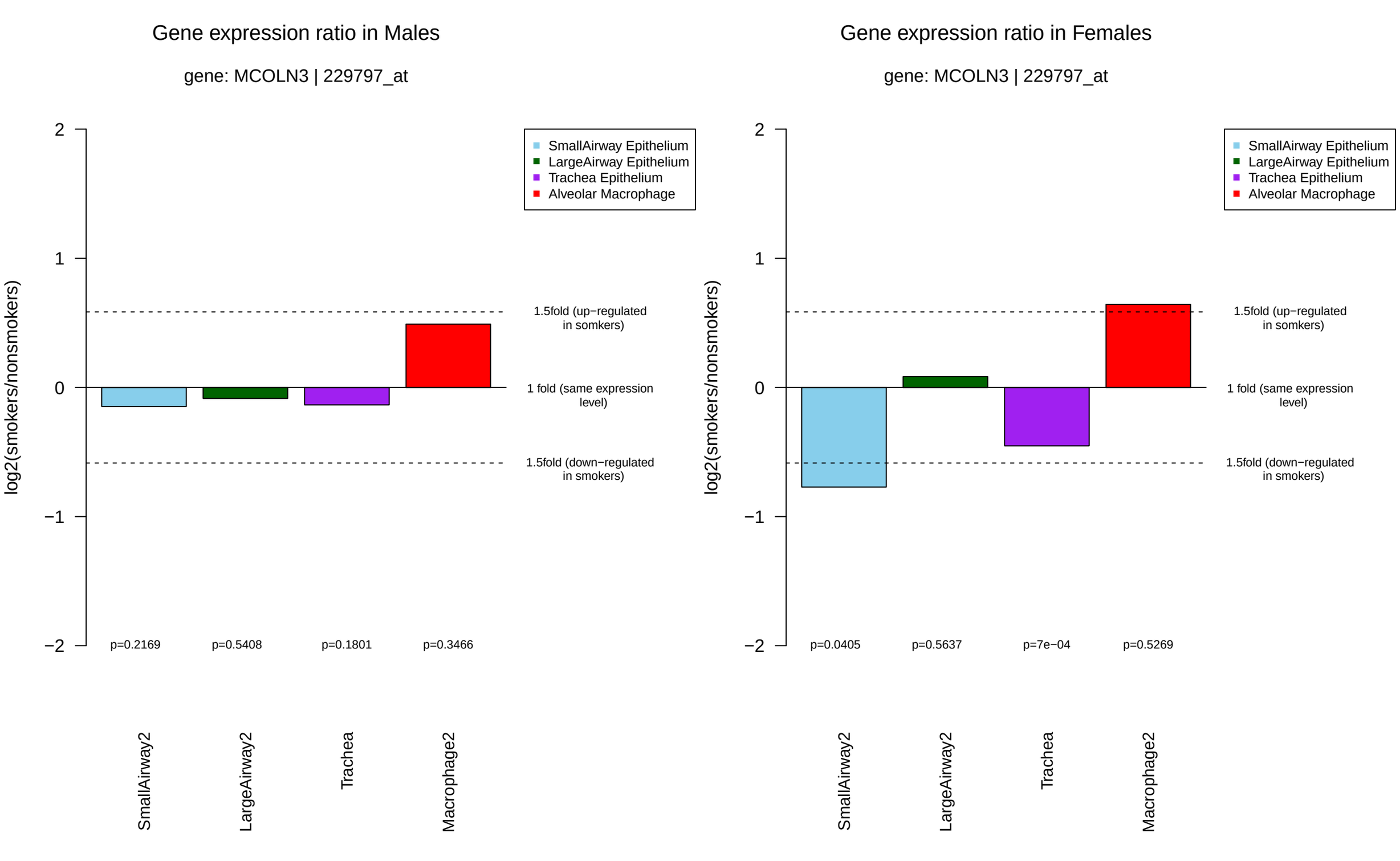
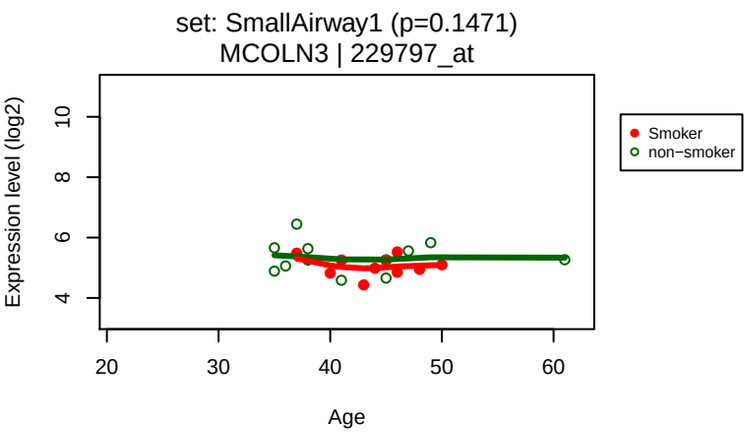
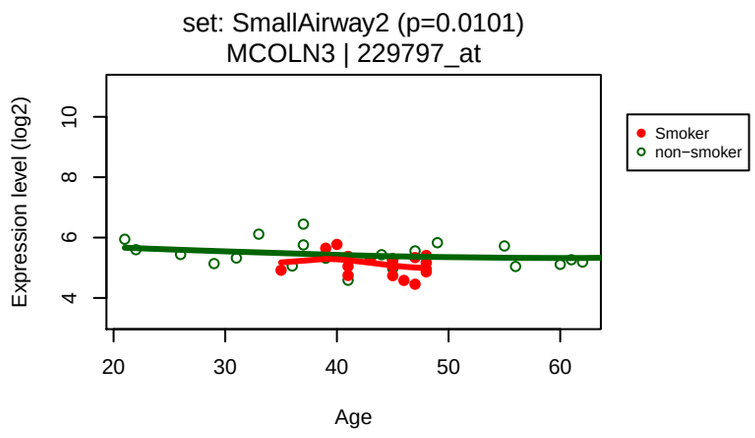
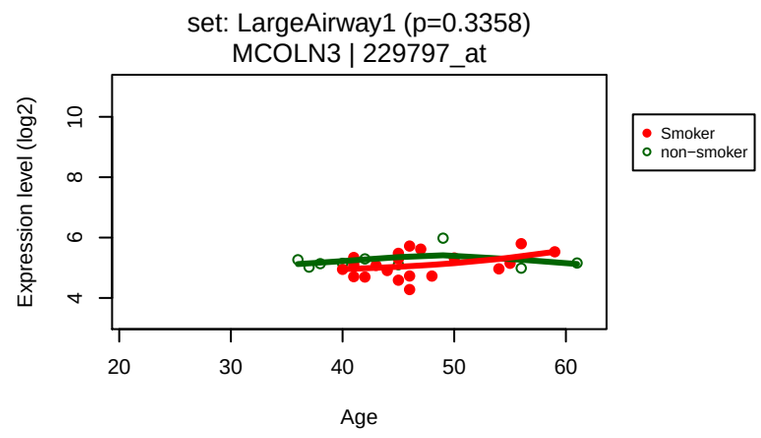

Gene expression of smokers and nonsmokers
gene: MCOLN3 | 229797_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1471
p=0.0101
p=0
p=0.3358
p=0.6547
p=0.025
p=0.0033
p=0.2096
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

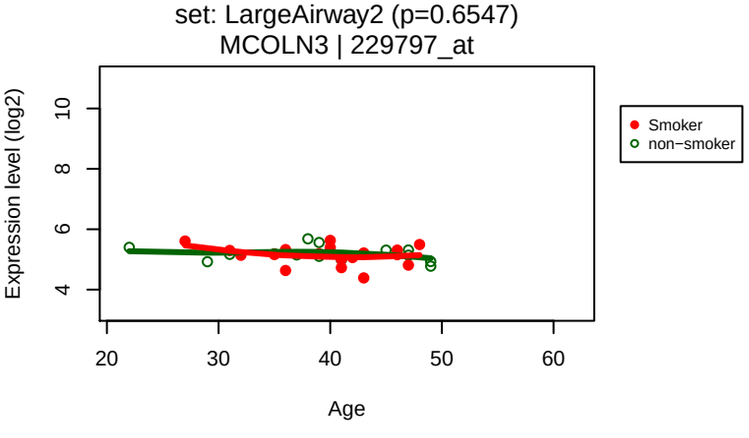

Gene expression ratio (smokers/nonsmokers)
gene: MCOLN3 | 229797_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1471
p=0.0101
p=0
p=0.3358
p=0.6547
p=0.025
p=0.0033
p=0.2096
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

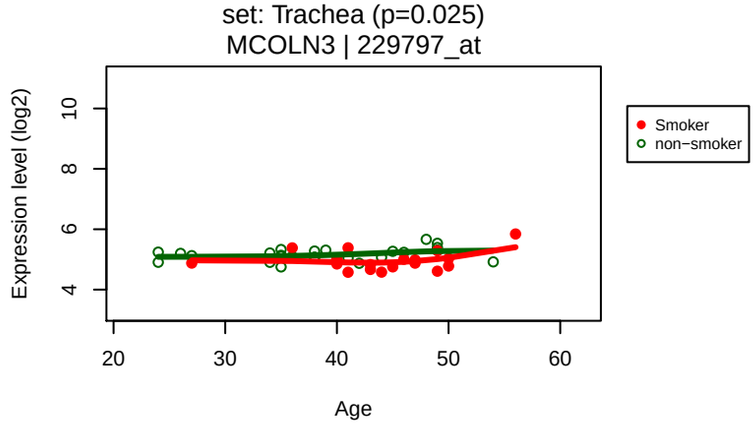

Gene expression ratio in Males
gene: MCOLN3 | 229797_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2169
p=0.5408
p=0.1801
p=0.3466
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCOLN3 | 229797_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0405
p=0.5637
p=7e−04
p=0.5269
SmallAirway2
LargeAirway2
Trachea
Macrophage2

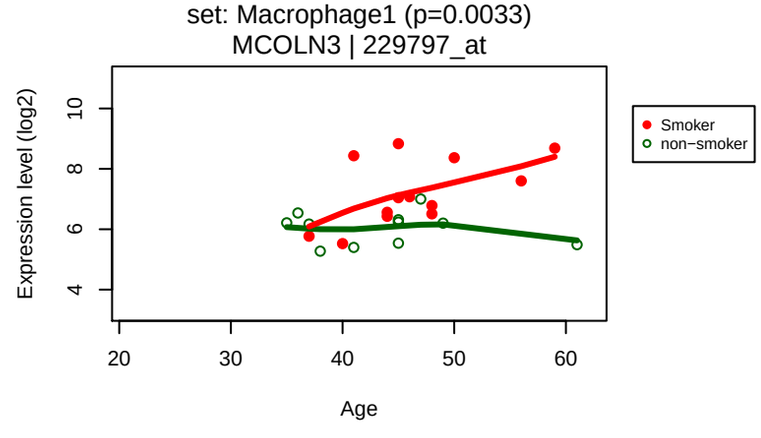

set: SmallAirway1 (p=0.1471)
MCOLN3 | 229797_at
Expression level (log2)
Age
Smoker
non−smoker

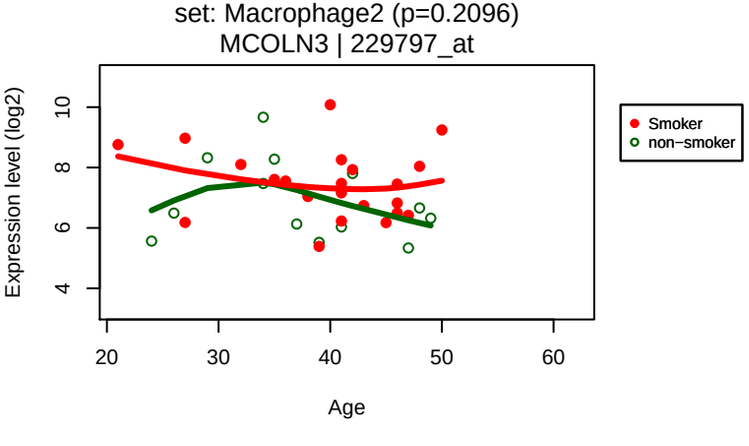

set: SmallAirway2 (p=0.0101)
MCOLN3 | 229797_at
Expression level (log2)
Age
Smoker
non−smoker

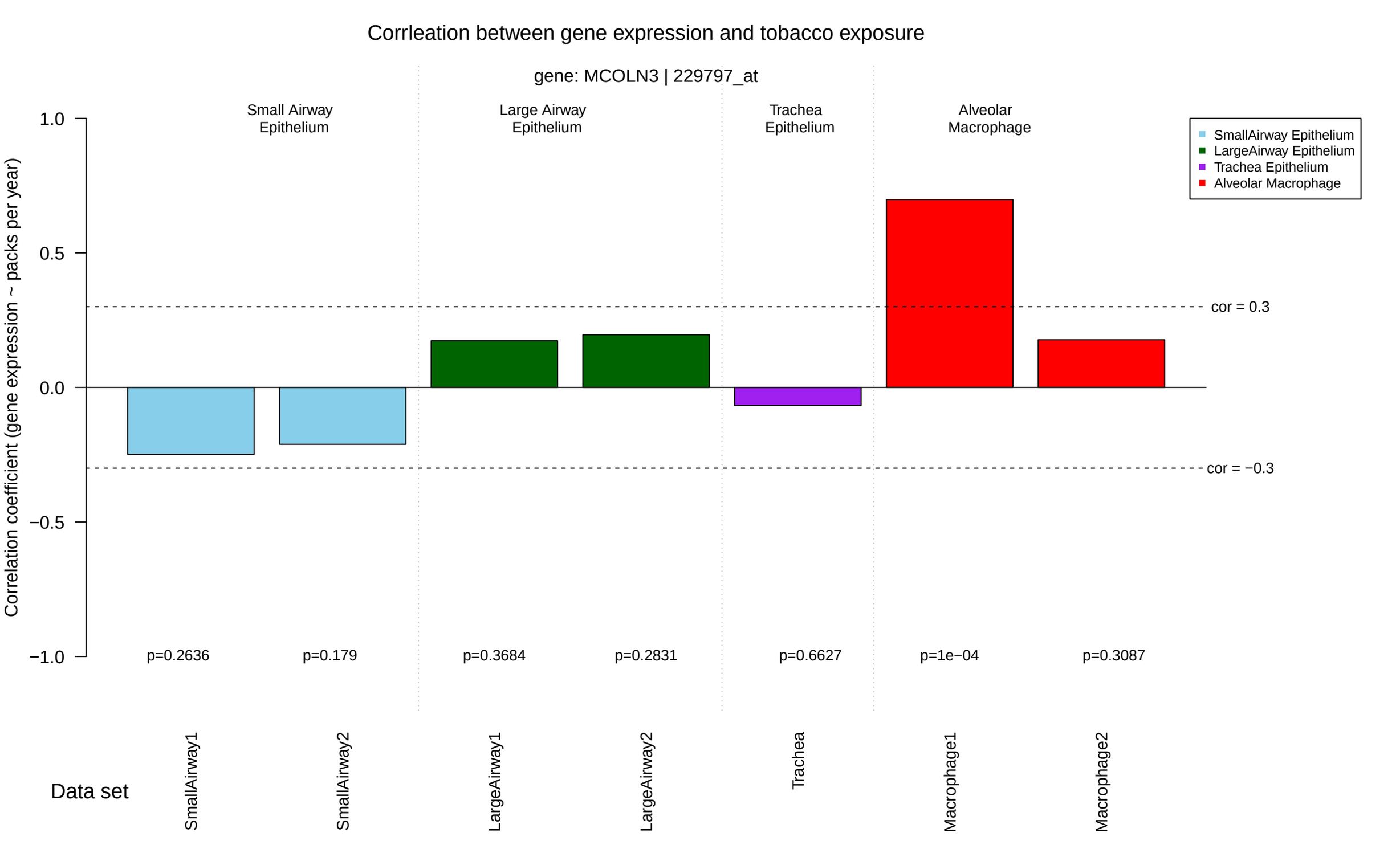

set: LargeAirway1 (p=0.3358)
MCOLN3 | 229797_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6547)
MCOLN3 | 229797_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.025)
MCOLN3 | 229797_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0033)
MCOLN3 | 229797_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2096)
MCOLN3 | 229797_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCOLN3 | 229797_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2636
p=0.179
p=0.3684
p=0.2831
p=0.6627
p=1e−04
p=0.3087
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2